



2017

HIGHER
SCHOOL
CERTIFICATE
TRIAL HSC
EXAMINATION

Mathematics Extension 2

Examiners Mr J. Dillon, Mr G. Huxley, Mr G. Rawson and Mrs D. Crancher

General

Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided for your use
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations

Total marks:
100

Section I – 10 marks (pages 2 – 5)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 6 – 14)

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section

Name: _____

Teacher: _____

Section I

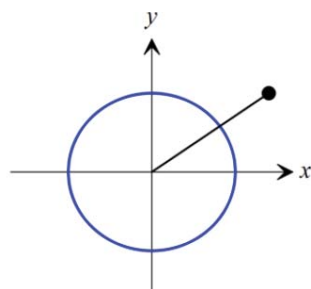
10 marks

Attempt Questions 1 and 10

Allow about 15 minutes for this section

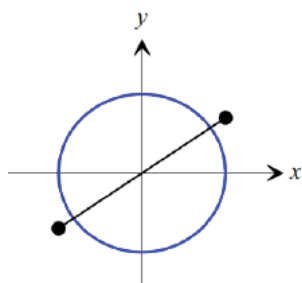
Use the multiple-choice answer sheet for Questions 1 – 10

1. The Argand diagram below shows the complex number z , represented by a vector, along with the unit circle.

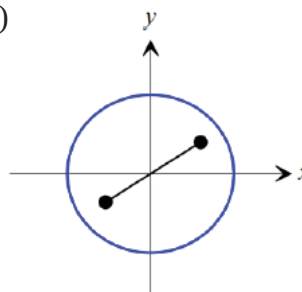


Which diagram best illustrates the vectors representing $\sqrt{z}$?

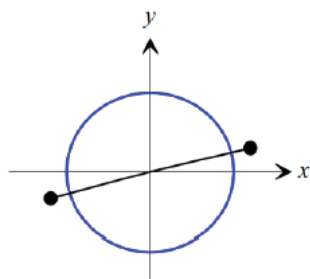
(A)



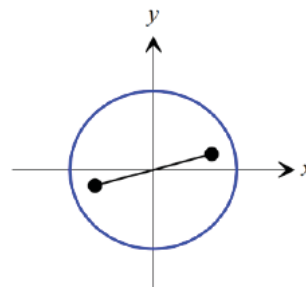
(B)



(C)



(D)



2. Which line intersects the circle $|z - 3 - 2i| = 2$ twice?
- (A) $|z - 3 - 2i| = |z - 5|$ (B) $|z - i| = |z + 1|$
- (C) $\operatorname{Re}(z) = 5$ (D) $\operatorname{Im}(z) = 0$
3. The polynomial equation $x^3 + x^2 - x - 4 = 0$ has roots α , β and γ . Which of the following polynomial equations has roots α^2 , β^2 and γ^2 ?
- (A) $x^3 - 3x^2 + 9x - 16 = 0$ (B) $x^3 - 3x^2 - 8x - 16 = 0$
- (C) $x^3 - x^2 + 9x - 1 = 0$ (D) $x^3 - x^2 - 8x - 1 = 0$
4. What are the values of real numbers p and q such that $1 - i$ is a root of the equation $z^3 + pz + q = 0$?
- (A) $p = -2$ and $q = -4$ (B) $p = -2$ and $q = 4$
- (C) $p = 2$ and $q = 4$ (D) $p = 2$ and $q = 4$
5. The equation $x^3 - y^3 + 3xy + 1 = 0$ defines y implicitly as a function of x .
What is the value of $\frac{dy}{dx}$ at the point $(1, 2)$?
- (A) $\frac{1}{3}$ (B) $\frac{1}{2}$
- (C) $\frac{3}{4}$ (D) 1

6. What is the natural domain of the function $f(x) = \frac{1}{2} \left(x\sqrt{x^2-1} - \ln(x + \sqrt{x^2-1}) \right)$?

(A) $x \leq -1$ or $x \geq 1$

(B) $-1 \leq x \leq 1$

(C) $x \geq 1$

(D) $x \leq -1$

7. The point $P(cp, \frac{c}{p})$ lies on the hyperbola $xy = c^2$

What is the equation of the normal to the hyperbola at P ?

(A) $p^2x - py + c - cp^4 = 0$

(B) $p^3x - py + c - cp^4 = 0$

(C) $x + p^2y - 2c = 0$

(D) $x + p^2y - 2cp = 0$

8. What are the co-ordinates of the foci of the graph of $xy = 12$?

(A) $(2\sqrt{3}, 2\sqrt{3})$ and $(-2\sqrt{3}, -2\sqrt{3})$

(B) $(2\sqrt{6}, 2\sqrt{6})$ and $(-2\sqrt{6}, -2\sqrt{6})$

(C) $(2\sqrt{3}, 0)$ and $(-2\sqrt{3}, 0)$

(D) $(2\sqrt{6}, 0)$ and $(-2\sqrt{6}, 0)$

9. The substitution of $x = \sin \theta$ in the integral $\int_0^{\frac{1}{2}} \frac{x^2}{\sqrt{1-x^2}} dx$ results in which integral?

(A) $\int_0^{\frac{1}{2}} \frac{\sin^2 \theta}{\cos \theta} d\theta$

(B) $\int_0^{\frac{1}{2}} \sin^2 \theta d\theta$

(C) $\int_0^{\frac{\pi}{6}} \frac{\sin^2 \theta}{\cos \theta} d\theta$

(D) $\int_0^{\frac{\pi}{6}} \sin^2 \theta d\theta$

- (D) 72

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question in a new answer booklet.

All necessary working should be shown in every question.

Question 11 (15 marks) Answer this question in a new answer booklet

- (a) (i) Simplify i^{2017} 1
- (ii) Sketch the locus of $\arg(z-1) = \frac{\pi}{4}$ 1
- (b) $z = -\sqrt{3} + i$ and $w = 1 + i$
- (i) Find $\frac{z}{w}$ in Cartesian form. 2
- (ii) Convert both z and w to modulus – argument form. 3
- (iii) Use your answers to (i) and (ii) to find the exact value of $\cos \frac{7\pi}{12}$. 1
- (c) $(x+iy)^2 = 7-24i$, where x and y are real.
- (i) Find the exact values of x and y . 3
- (ii) Hence, solve the equation $2z^2 + 6z + (1+12i) = 0$. 2
- (d) Use De Moivre's Theorem to show that $(\cot \theta + i)^n + (\cot \theta - i)^n = \frac{2 \cos n\theta}{\sin^n \theta}$. 2

Question 12 (15 marks) *Answer this question in a new answer booklet*

- (a) The equation $32x^3 - 16x^2 - 2x + 1 = 0$ has roots α , β , and γ .
- (i) What is the value of γ if $\gamma = \alpha + \beta$? 2
- (ii) Fully factorise $P(x) = 32x^3 - 16x^2 - 2x + 1$ 2
- (b) The polynomial $P(z) = z^4 - 5z^3 + az^2 + bz - 10$ where a and b are real.
- Given that $2+i$ is a zero of $P(z)$, write $P(z)$ as a product of two real quadratic factors. 2
- (c) $P(x) = x^4 + ax^2 + bx + 28$ has a double root at $x=2$.
- Find a and b . 2
- (d) When $P(x)$ is divided by $(x-2)$ and $(x+3)$ the respective remainders are -7 and 3 .
- Find the remainder when $P(x)$ is divided by $(x-2)(x+3)$. 2
- (e) Let $z=1+i$ be a root of: $z^2 - b iz + c = 0$, where b and c are real.
- (i) Find b and c 2
- (ii) Find the other root of the polynomial. 1
- (f) Solve the equation $x^4 - 5x^3 - 9x^2 + 81x - 108 = 0$ given that it has a triple root. 2

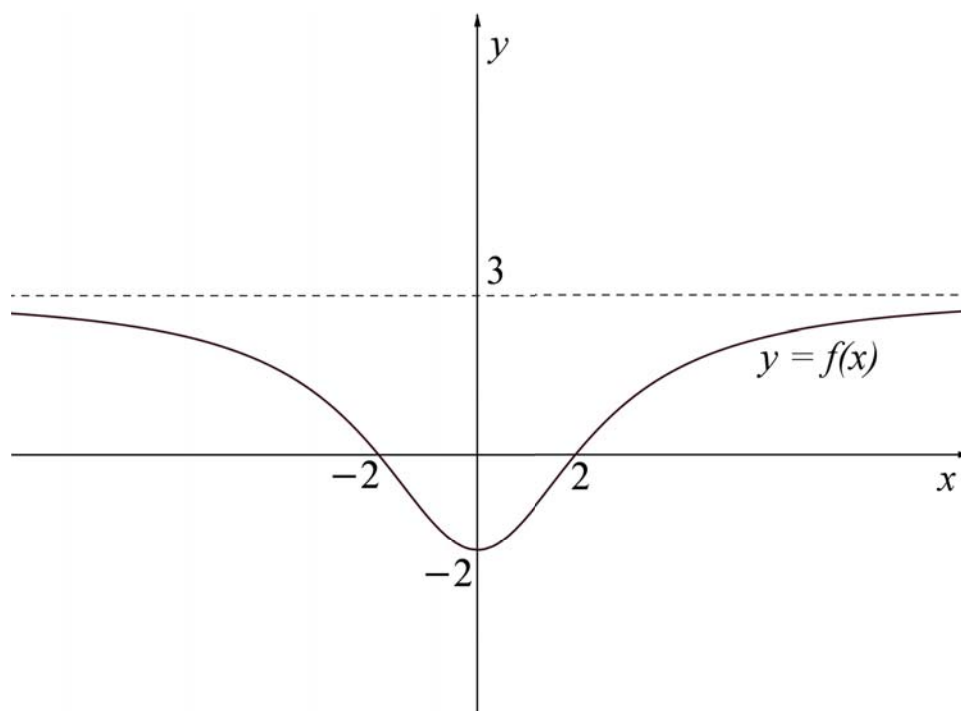
Question 13 (15 marks) *Answer this question in a new answer booklet*

- (a) (i) By writing $\frac{(x-2)(x-5)}{x-1}$ in the form $mx + b + \frac{a}{x-1}$, find the equation of the oblique asymptote of $y = \frac{(x-2)(x-5)}{x-1}$. 2
- (ii) Hence sketch the graph of $y = \frac{(x-2)(x-5)}{x-1}$, clearly indicating all intercepts and asymptotes. 2
- (b) Let $f(x) = 3x^5 - 10x^3 + 16x$
- (i) Show that $f'(x) \geq 1$ for all x . 2
- (ii) For what values of x is $f''(x)$ decreasing? 2
- (iii) Sketch the graph of $y = f(x)$, indicating any turning points and points of inflexion. 2

Question 13 continues on the next page

Question 13 continued

- (c) The diagram shows the graph of $y = f(x)$.



Draw separate one-third page sketches of the graphs of the following:

- | | | |
|-------|----------------|---|
| (i) | $y = f(x) $ | 1 |
| (ii) | $y = e^{f(x)}$ | 2 |
| (iii) | $y^2 = f(x)$ | 2 |

Question 14 (15 marks)*Answer this question in a new answer booklet*

- (a) $A(5\cos\theta, 4\sin\theta)$ is a point on the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$.

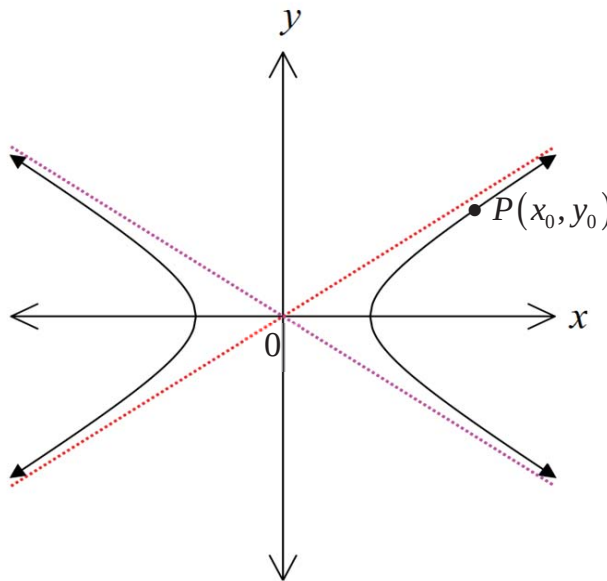
The normal at A meets the x -axis at P and the y -axis at Q .

- (i) Show that the normal to the ellipse at A has the equation

$$5x\sin\theta - 4y\cos\theta = 9\sin\theta\cos\theta \quad 2$$

- (ii) M is the midpoint of PQ . Show that the locus of M is an ellipse. 3

- (b) The point $P(x_0, y_0)$ lies on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, where $a > b > 0$.



- (i) Write the equations of the asymptotes to the hyperbola in general form. 1
- (ii) Write an expression for $\tan\theta$, where θ is the acute angle between the asymptotes, in terms of a and b . 2
- (iii) Hence, write an expression for $\sin\theta$. 1

Question 14 continues on the next page

Question 14 continued

- (iv) If C and D are the feet of the perpendiculars drawn from $P(x_0, y_0)$ to the asymptotes show that $CP \times DP = \frac{a^2 b^2}{a^2 + b^2}$ **3**
- (v) Prove that $OCPD$, where O is the origin, is a cyclic quadrilateral. **1**
- (vi) Calculate the area of $\triangle PCD$. **2**

Question 15 (15 marks) *Answer this question in a new answer booklet*

(a) Find $\int \frac{x^2}{(1-x^2)^{\frac{3}{2}}} dx$ using the substitution $x = \sin \theta$ with $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. 2

(b) Use the substitution $t = \tan \frac{x}{2}$ to show that $\int_0^{\frac{\pi}{2}} \frac{1}{3 - \cos x - 2 \sin x} dx = \frac{\pi}{2}$. 4

(c) (i) Find the real numbers a , b and c such that 3

$$\frac{7x+4}{(x^2+1)(x+2)} \equiv \frac{ax+b}{x^2+1} + \frac{c}{x+2}$$

(ii) Hence, find $\int \frac{7x+4}{(x^2+1)(x+2)} dx$ 2

(d) (i) Let $I_n = \int_0^{\frac{\pi}{2}} \cos^n t dt$. 2

Show that $I_n = \left(\frac{n-1}{n} \right) I_{n-2}$ with $n \geq 2$.

(ii) Hence, otherwise, show that the exact value of $\int_0^{\frac{\pi}{2}} \cos^4 t dt = \frac{3\pi}{16}$. 2

Question 16 (15 marks) *Answer this question in a new answer booklet*

- (a) A School Council consists of six year 12 students and five year 11 students, from whom a committee of five members is chosen at random.

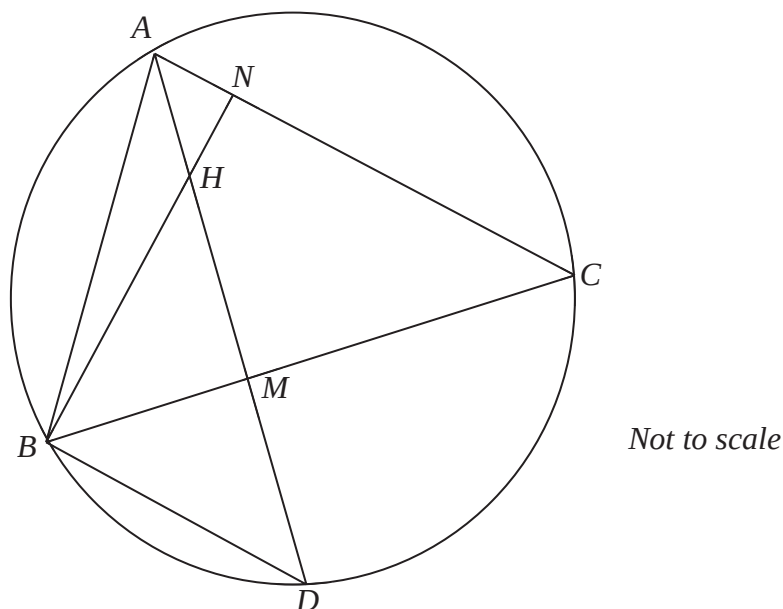
What is the probability that the year 12 students have a majority on the committee?

2

- (b) In the circle below, points A , B and C lie on the circumference of a circle. The altitudes AM and BN of an acute angled triangle ABC meet at H . AM produced cuts the circle at D .

Prove that $HM = MD$.

3



- (c) The n th Fermat number, F_n , is defined by $F_n = 2^{2^n} + 1$ for $n = 0, 1, 2, 3, \dots$.

Prove by mathematical induction, that for all positive integers:

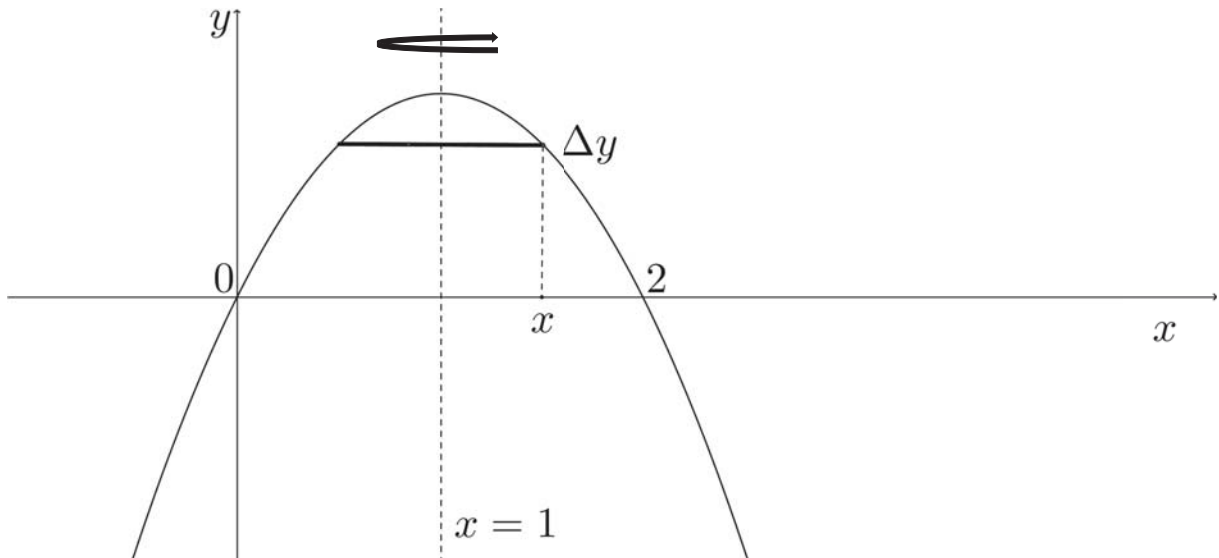
$$F_0 \times F_1 \times F_2 \times \dots \times F_{n-1} = F_n - 2$$

4

Question 16 continues on the next page

Question 16 continued

- (d) The area bounded by the curve $y = 2x - x^2$ and the x -axis is rotated through 180° about the line $x = 1$.



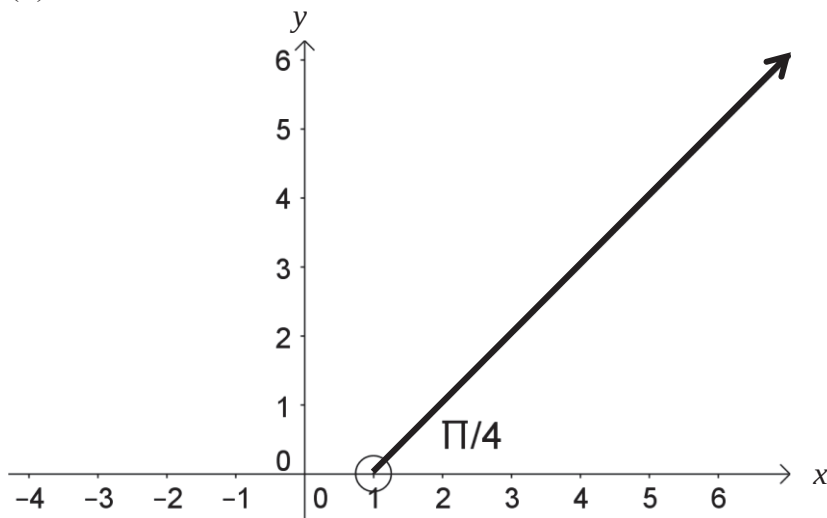
- (i) Show that the volume, ΔV , of a representative horizontal slice of width Δy is given by 2

$$\Delta V = \pi(x-1)^2 \Delta y$$

- (ii) Hence, show that the volume of the solid of revolution is given by 2

$$V = \lim_{\Delta y \rightarrow 0} \sum_{y=0}^1 \pi(1-y) \Delta y$$

- (iii) Hence, find the volume of the solid of revolution. 2

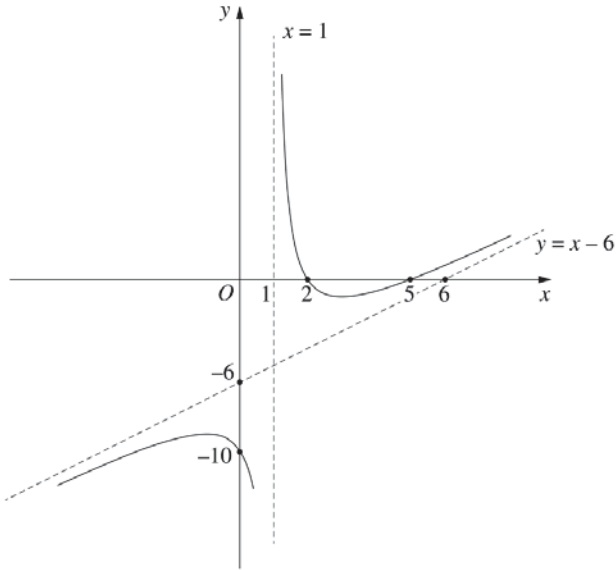
Year 12 2017		Mathematics Extension 2	Task 4 Trial
Question No. 11		Solutions and Marking Guidelines	
Outcomes Addressed in this Question			
E3 uses the relationship between algebraic and geometric representations of complex numbers and of conic sections			
Outcome	Solutions		Marking Guidelines
E3	(a) (i) $i^{2017} = (i^{2016})(i)$ $= (i^4)^{504} (i)$ $= i$		1 mark for correct solution
E3	(ii) 		1 mark for correct diagram
E3	(b) (i) $\frac{z}{w} = \frac{-\sqrt{3}+i}{1+i} \times \frac{1-i}{1-i}$ $= \frac{-\sqrt{3}+i\sqrt{3}+i+1}{2}$ $= \left(\frac{1-\sqrt{3}}{2}\right) + i\left(\frac{1+\sqrt{3}}{2}\right)$		2 marks for complete correct solution 1 mark for substantial working that could lead to a correct solution

E3	(ii) $z = -\sqrt{3} + 1$ $ z = \sqrt{(-\sqrt{3})^2 + (1)^2}$ $= \sqrt{3+1}$ $= 2$ $\text{Arg}(z) = \tan^{-1}\left(\frac{1}{-\sqrt{3}}\right)$ $= \frac{5\pi}{6}$ $\therefore z = 2\text{cis}\frac{5\pi}{6}$	3 marks for complete correct solution 2 marks for substantial working that could lead to a correct solution with only one error 1 mark for substantial working that could lead to a correct solution
E3	(iii) $\frac{z}{w} = \frac{2\text{cis}\frac{5\pi}{6}}{\sqrt{2}\text{cis}\frac{\pi}{4}}$ $= \sqrt{2}\text{cis}\frac{7\pi}{12}$ $= \sqrt{2} \cos \frac{7\pi}{12} + i \left(\sqrt{2} \sin \frac{7\pi}{12} \right)$ Equating real parts: $\sqrt{2} \cos \frac{7\pi}{12} = \frac{1-\sqrt{3}}{2}$ $\therefore \cos \frac{7\pi}{12} = \frac{1-\sqrt{3}}{2\sqrt{2}} \quad \text{or} \quad \frac{\sqrt{2}-\sqrt{6}}{4}$	1 mark for complete correct solution

E3	(c) (i) $(x + iy)^2 = 7 - 24i$ $x^2 + 2ixy - y^2 = 7 - 24i$ $x^2 - y^2 = 7 \dots\dots\dots(1)$ $2xy = -24$ $\therefore y = \frac{-12}{x} \dots\dots\dots(2)$ sub (2) into (1): $x^2 - \left(\frac{-12}{x}\right)^2 = 7$ $x^2 - \frac{144}{x^2} = 7$ $x^4 - 144 = 7x^2$ $x^4 - 7x^2 - 144 = 0$ $(x^2 - 16)(x^2 + 9) = 0$ $\therefore x = \pm 4$ as x is real If $x = 4$, $y = \frac{-12}{4} = -3$ $x = -4$, $y = 3$ Therefore solutions for $(x + iy)^2 = 7 - 24i$ are $x = 4$, $y = -3$ and $x = -4$, $y = 3$.	3 marks for complete correct solution 2 marks for substantial working that could lead to a correct solution with only one error 1 mark for substantial working that could lead to a correct solution
E3	(ii) $2z^2 + 6z + (1 + 12i) = 0$ $\Delta = b^2 - 4ac$ $= 6^2 - 4 \times 2(1 + 12i)$ $= 28 - 96i$ $= 4(7 - 24i)$ $\therefore z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $= \frac{-6 \pm \sqrt{4(7 - 24i)}}{2 \times 2}$ $= \frac{-6 \pm 2(4 - 3i)}{2 \times 2}$ $= \frac{1 - 3i}{2}, \frac{-7 + 3i}{2}$	2 marks for complete correct solution 1 mark for substantial work that could lead to a correct solution

E3	(d) $ \begin{aligned} & (\cot \theta + i)^n + (\cot \theta - i)^n \\ &= \left(\frac{\cos \theta + i \sin \theta}{\sin \theta} \right)^n + \left(\frac{\cos \theta - i \sin \theta}{\sin \theta} \right)^n \\ &= \frac{1}{\sin^n \theta} \left\{ (\cos \theta + i \sin \theta)^n + (\cos(-\theta) + i \sin(-\theta))^n \right\} \\ &= \frac{1}{\sin^n \theta} (\cos n\theta + i \sin n\theta + \cos(-n\theta) + i \sin(-n\theta)) \quad \text{using de Moivre's theorem} \\ &= \frac{1}{\sin^n \theta} (\cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta) \\ &= \frac{2 \cos n\theta}{\sin^n \theta} \end{aligned} $ Multiple Choice Answers: 1. C 2. A 3. A 4. B 5. D 6. C 7. B 8. B 9. D 10. C	2 marks for complete correct solution 1 mark for substantial work that could lead to a correct solution
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Year 12	Mathematics Extension 2	Task 4 (Trial) 2017
Question 12	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
E4	uses efficient techniques for the algebraic manipulation required in dealing with questions involving polynomials	
Part	Solutions	Marking Guidelines
(a) (i)	Roots are $\alpha, \beta, \alpha + \beta$. $\Sigma \alpha = 2\alpha + 2\beta = -\left(\frac{-16}{32}\right) = \frac{1}{2} \quad \therefore \alpha + \beta = \delta = \frac{1}{4}$	(a)(i) 2 Marks ~ Correct solution. 1 Mark ~ Makes significant progress towards the solution (Uses correct sum)
(ii)	Test $P\left(\frac{1}{4}\right) = 0$ $(4x-1)$ is a factor. The other two roots have $\alpha + \beta = \frac{1}{4}$; $\alpha\beta = \frac{-1}{8}$ from sum and product of roots of $P(x)$ $\therefore a\left(x^2 - \frac{1}{4}x - \frac{1}{8}\right) = 0$ $8x^2 - 2x - 1 = 0$ $\therefore P(x) = (4x-1)(4x+1)(2x-1)$	(a) (ii) 2 Marks ~ Correct factorisation. 1 Mark ~ Makes significant progress towards solution
(b)	Real coefficients. So $2-i$ is also a root. $2+i+2-i=4 \quad (2+i)(2-i)=5$ So one quadratic factor is: $(z^2 - 4z + 5)$ If the other 2 roots are α, β : $4 + \alpha + \beta = 5 \rightarrow \alpha + \beta = 1$ $5\alpha\beta = -10 \rightarrow \alpha\beta = -2$ $\therefore P(z) = (z^2 - 4z + 5)(z^2 - z - 2)$	(b) 2 Marks ~ Correct solution. Must be quadratic factors. 1 Mark ~ Makes significant progress towards the solution (One correct quadratic factor)
(c)	$P(2) = 0 \rightarrow 4a + 2b = -44$ $P'(2) = 0 \rightarrow 4a + b = -32 \quad \therefore a = -5 \quad b = -12$	(c) 2 marks: Correct solution 1 Mark ~ Significant progress towards solution.
(d)	$P(x) = Q(x)(x-2)(x+3) + ax + b$ $P(2) = 2a + b = -7$ $P(-3) = -3a + b = 3 \quad \therefore a = -2 \quad b = -3$ Remainder = $-2x - 3$	(d) 2 Marks ~ Correct remainder . 1 Marks ~ Makes significant progress towards the solution.
(e) (i)	$(1+i)^2 - bi(1+i) + c = 0$ $(2-b)i + b + c = 0i + 0. \quad \therefore b = 2 \quad c = -2$	(e) (i) 2 Marks ~ Correct solution. 1 Mark ~ Makes significant progress towards solution(substitution completed.)
(ii)	Sum of roots = $bi = 2i$ So, other roots = $2i - (1+i) = i - 1$	(e) (ii) 1 mark: correct answer.
(f)	$P(x) = x^4 - 5x^3 - 9x^2 + 81x - 108$ $P'(x) = 4x^3 - 15x^2 - 18x + 81$ $P''(x) = 12x^2 - 30x - 18 = 6(2x+1)(x-3)$ Possible triple roots are $x=3, -\frac{1}{2}$ $P\left(-\frac{1}{2}\right) \neq 0; P(3) = P'(3) = P''(3) = 0$ Product of roots = -108 , so solutions are: $x=3, -4$	(f) 2 marks: Correct roots with justification. 1 mark: Significant progress towards correct solution.

Year 12	Mathematics Extension 2	Trial Exam 2017 HSC
Question No. 13	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
E6 - combines the ideas of algebra and calculus to determine the important features of the graphs of a wide variety of functions		
Part	Solutions	Marking Guidelines
(a)	(i) $\frac{(x-2)(x-5)}{x-1} = \frac{x^2 - 7x + 10}{x-1}$ $\begin{array}{r} x-6 \\ x-1 \overline{) x^2 - 7x + 10} \\ \underline{x^2 - x} \\ -6x + 10 \\ \underline{-6x + 6} \\ 4 \end{array}$ $\therefore \lim_{x \rightarrow \infty} \frac{(x-2)(x-5)}{x-1}$ $= \lim_{x \rightarrow \infty} x - 6 + \frac{4}{x-1}$ $= x - 6 + 0$ ie the oblique asymptote is $y = x - 6$ (ii) 	2 marks : correct solution 1 mark : substantially correct solution
	(b) (i) $f(x) = 3x^5 - 10x^3 + 16x$ $f'(x) = 15x^4 - 30x^2 + 16$ $= 15(x^2 - 1)^2 + 1$ $\geq 1, \text{ since } 15(x^2 - 1)^2 \geq 0 \text{ for all } x.$	2 marks : correct solution 1 mark : substantially correct solution

(ii) $f''(x)$ is decreasing when $f'''(x) < 0$

$$f''(x) = 60x^3 - 60x$$

$$f'''(x) = 180x^2 - 60 < 0$$

$$x^2 < \frac{1}{3}$$

$$-\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}}$$

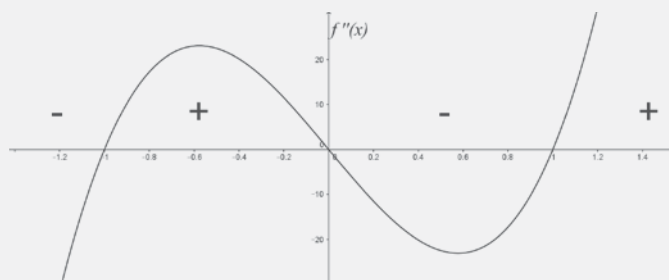
2 marks : correct solution

1 mark : substantially correct solution
(or reading "decreasing" as "negative"... which would give this solution...)

(ii) $f''(x) = 60x^3 - 60x$

$$= 60x(x^2 - 1)$$

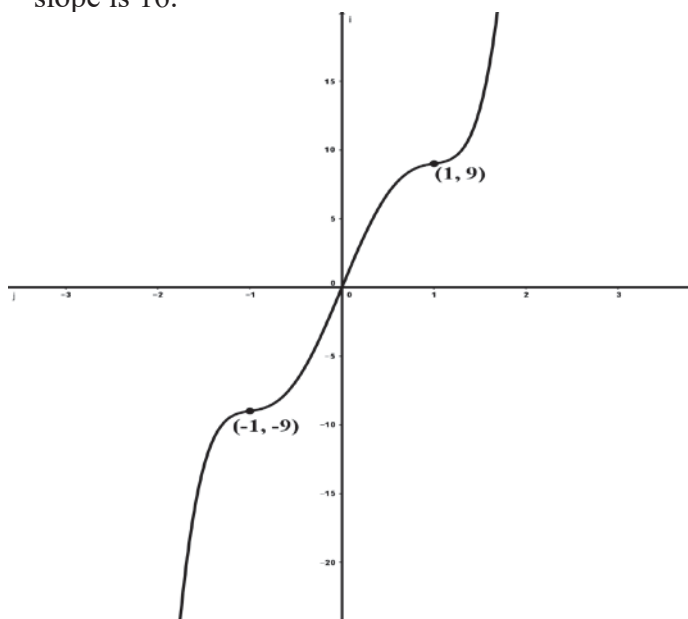
$$= 60x(x-1)(x+1)$$



$$f''(x) < 0 \text{ for } x < -1 \text{ and } 0 < x < 1$$

...which is wrong)

(iii) $f'(x) \geq 1$, so $f(x)$ is monotonic increasing, has no stationary points, and the smallest slope occurs at $x = \pm 1$ where $f'(x) = 1$, ie $(-1, -9)$ & $(1, 9)$. These are points of inflexion. the other point of inflexion is $(0, 0)$ where the slope is 16.

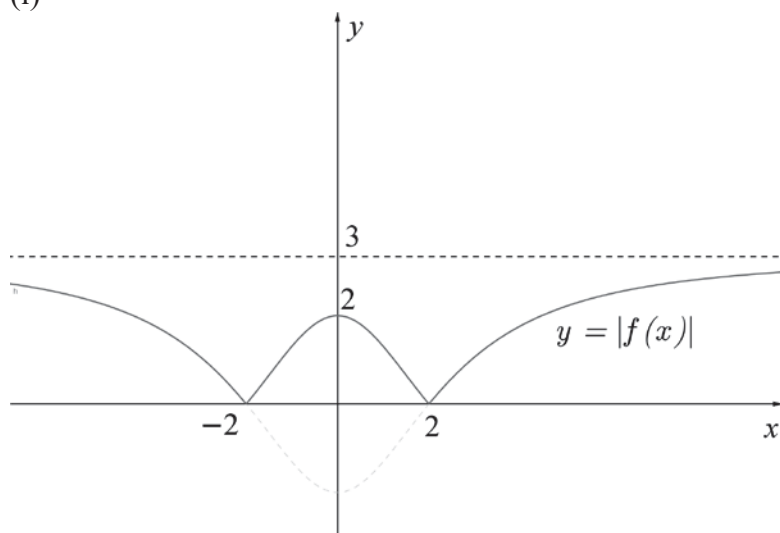


2 marks : correct solution

1 mark : substantially correct solution

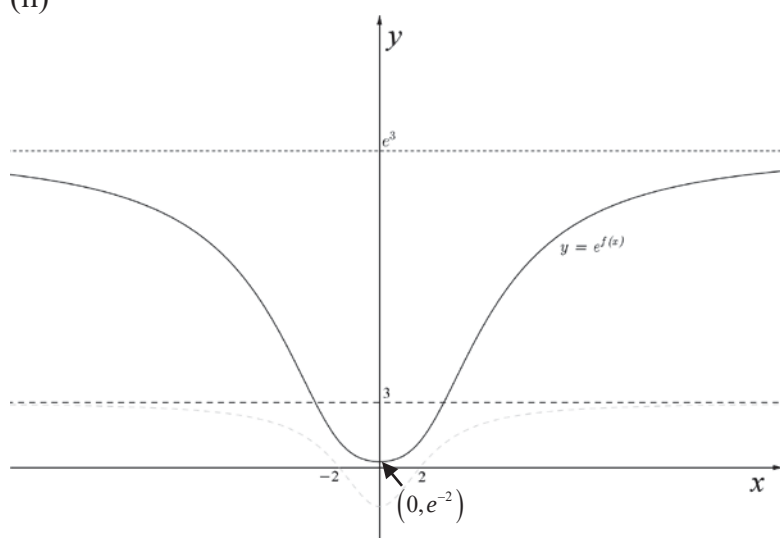
(c)

(i)



1 mark : correct solution

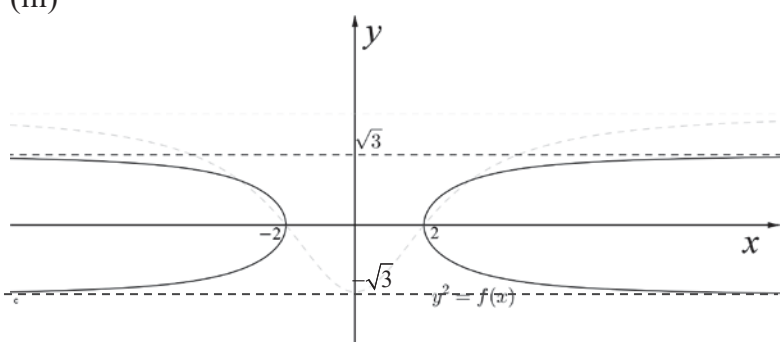
(ii)



2 marks : correct solution

1 mark : substantially correct solution

(iii)



2 marks : correct solution

1 mark : substantially correct solution

Year 12	Mathematics Extension 2	Task 4 (Trial) 2017
Question 14	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
E3	uses the relationship between algebraic and geometric representations of conic sections	
Part	Solutions	Marking Guidelines
(a) (i)	$\frac{d}{dx}\left(\frac{x^2}{25} + \frac{y^2}{16}\right) = \frac{d}{dx}(1) \quad \rightarrow \quad \frac{dy}{dx} = \frac{-16x}{25y}$	(a)(i) 2 Marks ~ Correct with working.
(ii)	Gradient of normal = $\frac{25y}{16x} = \frac{5 \sin \theta}{4 \cos \theta}$ $y - 4 \sin \theta = \frac{5 \sin \theta}{4 \cos \theta} (x - 5 \cos \theta)$ $4y \cos \theta - 16 \sin \theta \cos \theta = 5x \sin \theta - 25 \sin \theta \cos \theta$ $P = \left(\frac{9 \cos \theta}{5}, 0\right); Q = \left(0, \frac{-9 \sin \theta}{4}\right); M = \left(\frac{9 \cos \theta}{10}, \frac{-9 \sin \theta}{8}\right)$ Justify locus of M is an ellipse by eliminating $\sin \theta, \cos \theta$ and showing the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Alternately, it was accepted that M is in parametric form $(a \cos \theta, b \sin \theta)$.	1 Marks ~ Makes significant progress towards the solution (a)(ii) 3 marks: Finds P, Q, M and justifies the locus of M. 2 marks: Significant progress. 1 mark: Some relevant progress.
(b) (i)	$bx - ay = 0 \quad bx + ay = 0$	(b)(i) 1 mark : Correct answer in general form. If equations are correct but not in general form, you received this mark if you used general form in part (iv).
(ii)	$\tan \frac{\theta}{2} = \frac{b}{a} \quad \therefore \tan \theta = \frac{2\left(\frac{b}{a}\right)}{1 - \left(\frac{b}{a}\right)^2} = \frac{2ab}{a^2 - b^2}$	(b)(ii) 2 marks : correct solution. 1 Mark ~ Makes significant progress towards solution
(iii)	$\sin \theta = \frac{2\left(\frac{b}{a}\right)}{1 + \left(\frac{b}{a}\right)^2} = \frac{2ab}{a^2 + b^2}$	<i>Note: Many people used angle between 2 lines formula, but it was easier to use double angle result which becomes a form of the t result because that led to more easily achieving part (iii)</i>
(iv)	CP and DP are perpendicular distances from P to the asymptotes. $CP \times DP = \frac{ bx_0 - ay_0 }{\sqrt{(-a)^2 + b^2}} \times \frac{ bx_0 + ay_0 }{\sqrt{a^2 + b^2}}$ $= \frac{(bx_0)^2 - (ay_0)^2}{a^2 + b^2} = \frac{b^2 x_0^2 - a^2 y_0^2}{a^2 + b^2}$ $= \frac{a^2 b^2}{a^2 + b^2}$	(b)(iii) 1 mark : Correct answer. (b)(iv) 3 marks : Correct solution, realising that this is a "show" question. 2 marks : Significant progress.. 1 mark : Some relevant progress made.
(v)	$\angle OCP = \angle ODP = 90^\circ$ since CP, DP are perpendiculars $OCPD$ is cyclic because these angles are opposite and supplementary.	(b)(v) 1 mark : Indicating which angles are right angles, as well as giving the reason for being cyclic.
(vi)	$\Delta PCD = \frac{1}{2} \cdot CP \cdot PD \times \sin(180^\circ - \theta)$ $= \frac{1}{2} \times \frac{a^2 b^2}{a^2 + b^2} \times \frac{2ab}{a^2 + b^2} = \frac{a^3 b^3}{(a^2 + b^2)^2}$	(b)(vi) 2 marks : Correct solution, including showing the use of sin ratio of supplementary angle. 1 mark : Significant progress.

Year 12	Mathematics Extension 2	Task 4 (Trial HSC) 2017
Question 15	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
E8	applies further techniques of integration, including partial fractions, integration by parts and recurrence formulae, to problems	
Part	Solutions	Marking Guidelines
(a)	Let $x = \sin \theta$, $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$ $dx = \cos \theta d\theta$ $(1-x^2)^{\frac{3}{2}} = (1-\sin^2 \theta)^{\frac{3}{2}}$ $= (\cos^2 \theta)^{\frac{3}{2}} = \cos^3 \theta$ $\int \frac{x^2}{(1-x^2)^{\frac{3}{2}}} dx = \int \frac{\sin^2 \theta}{\cos^3 \theta} \cos \theta d\theta$ $= \int \tan^2 \theta d\theta = \int (\sec^2 \theta - 1) d\theta$ $= \tan \theta - \theta + c = \frac{x}{(1-x^2)^{\frac{1}{2}}} - \sin^{-1} x + c$	2 Marks ~ Correct solution. 1 Marks ~ Makes significant progress towards the solution
(b)	$t = \tan \frac{x}{2}$ $\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2} \text{ or } dx = \frac{2}{1+t^2} dt$ When $x=0$ then $t=0$ and when $x=\frac{\pi}{2}$ then $t=1$ $3 - \cos x - 2 \sin x = \frac{3(1+t^2) - (1-t^2) - 4t}{1+t^2}$ $= \frac{3+3t^2-1+t^2-4t}{1+t^2}$ $= \frac{2(2t^2-2t+1)}{1+t^2}$ $= 2 \left[\left(t - \frac{1}{2}\right)^2 + \frac{1}{4} \right] \frac{2}{1+t^2}$ $\int_0^{\frac{\pi}{2}} \frac{1}{3 - \cos x - 2 \sin x} dx = \int_0^1 \frac{1}{2 \left[\frac{1}{\left(t - \frac{1}{2}\right)^2 + \frac{1}{4}} \right]} \times \frac{1+t^2}{2} \times \frac{2}{1+t^2} dt$ $= \int_0^1 \frac{1}{2 \left[\frac{1}{\left(t - \frac{1}{2}\right)^2 + \frac{1}{4}} \right]} dt \quad \left[\text{Let } u = t - \frac{1}{2}, du = dt \right]$ $= \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{2 \left[\frac{1}{u^2 + \frac{1}{4}} \right]} du$ $= \frac{1}{2} \left[2 \tan^{-1} u \right]_{-\frac{1}{2}}^{\frac{1}{2}}$ $= \tan^{-1} 1 - \tan^{-1} (-1)$ $= \frac{\pi}{2}$	4 Marks ~ Correct answer 3 Marks ~ Correctly determines the primitive function (in terms of t or another variable). 2 Marks ~ Correctly expresses the integral in terms of t. 1 Mark ~ Correctly finds dx in terms of dt and determines the new limits.

(c) (i)	$\frac{7x+4}{(x^2+1)(x+2)} = \frac{ax+b}{x^2+1} + \frac{c}{x+2}$ $7x+4 = (ax+b)(x+2) + c(x^2+1)$ Let $x = -2$ and $x = 0$ $-10 = 5c \qquad 4 = b(0+2) - 2(0^2+1)$ $c = -2 \qquad b = 3$ Equating the coefficients of x^2 $0 = a - 2$ or $a = 2$ $\therefore a = 2, b = 3$ and $c = -2$	3 Marks ~ Correct answer. 2 Marks ~ Calculates two of the variables 1 Mark ~ Makes some progress in finding a, b or c.
(ii)	$\int \frac{7x+4}{(x^2+1)(x+2)} dx = \int \frac{2x+3}{x^2+1} - \frac{2}{x+2} dx$ $= \int \frac{2x}{x^2+1} + \frac{3}{x^2+1} - \frac{2}{x+2} dx$ $= \ln x^2+1 + 3 \tan^{-1} x - 2 \ln x+2 + c$ $= \ln \left \frac{x^2+1}{(x+2)^2} \right + 3 \tan^{-1} x + c$	2 Marks ~ Correct answer. 1 Mark ~ Correctly finds one of the integrals.
(d) (i)	$I_n = \int_0^x \cos^n x dx$ $= \int_0^{\frac{\pi}{2}} \cos^n x dx$ Integration by parts $I_n = \int_0^{\frac{\pi}{2}} \cos^{n-1} t \cos t dt$ $= \left[\cos^{n-1} t \sin t \right]_0^{\frac{\pi}{2}} + (n-1) \int_0^{\frac{\pi}{2}} \cos^{n-2} t \sin^2 t dt$ $= (n-1) \int_0^{\frac{\pi}{2}} \cos^{n-2} t \sin^2 t dt$ $= (n-1) \int_0^{\frac{\pi}{2}} \cos^{n-2} t (1 - \cos^2 t) dt$ $= (n-1) \int_0^{\frac{\pi}{2}} (\cos^{n-2} t - \cos^n t) dt$ $= (n-1) \int_0^{\frac{\pi}{2}} \cos^{n-2} t dt - (n-1) \int_0^{\frac{\pi}{2}} \cos^n t dt$ Using the original integral $\int_0^{\frac{\pi}{2}} \cos^n t dt = (n-1) \int_0^{\frac{\pi}{2}} \cos^{n-2} t dt - n \int_0^{\frac{\pi}{2}} \cos^n t dt + \int_0^{\frac{\pi}{2}} \cos^n t dt$ $n \int_0^{\frac{\pi}{2}} \cos^n t dt = (n-1) \int_0^{\frac{\pi}{2}} \cos^{n-2} t dt$ $\int_0^{\frac{\pi}{2}} \cos^n t dt = \frac{(n-1)}{n} \int_0^{\frac{\pi}{2}} \cos^{n-2} t dt$ $I_n = \frac{(n-1)}{n} I_{n-2}$	2 Marks ~ Correct answer. 1 Mark ~ Correctly integrates by parts.

(ii)

$$I_n = \frac{(n-1)}{n} I_{n-2}$$

$$I_4 = \frac{(4-1)}{4} I_2$$

$$= \frac{3}{4} \int_0^{\frac{\pi}{2}} \cos^2 t dt$$

$$= \frac{3}{4} \int_0^{\frac{\pi}{2}} \frac{1}{2} (1 + \cos 2t) dt$$

$$= \frac{3}{8} \left[x + \frac{\sin 2t}{2} \right]_0^{\frac{\pi}{2}}$$

$$= \frac{3}{8} \left[\left(\frac{\pi}{2} + \frac{\sin \pi}{2} \right) - \left(0 + \frac{\sin 0}{2} \right) \right]$$

$$= \frac{3\pi}{16}$$

2 Marks ~ Correct answer.

1 Mark ~ Using the result from (d)(i) to obtain the definite integral.

Year 12 2017		Mathematics Extension 2	Task 4 Trial
Question No. 16		Solutions and Marking Guidelines	
Outcomes Addressed in this Question			
E2 chooses appropriate strategies to construct arguments and proofs in both concrete and abstract settings			
E7 uses the techniques of slicing and cylindrical shells to determine volumes			
E9 communicates abstract ideas and relationships using appropriate notation and logical argument			
Outcome	Solutions	Marking Guidelines	
E9	(a) Total arrangements = ${}^{11}C_5 = 462$ Majority of year 12 = $\left({}^6C_3 \times {}^5C_2\right) + \left({}^5C_1 \times {}^6C_4\right) + {}^6C_5 = 281$ Probability = $\frac{281}{462}$	2 marks for complete correct solution 1 marks for finding Total arrangements	
E2	(b) In $\triangle AMC$ and $\triangle ANH$ $\angle AMC = \angle ANH = 90^\circ$ ($AM \perp BC$ and $BN \perp AC$ since AM and BN are altitudes of $\triangle AMC$ and $\triangle ANH$) $\angle MAC = \angle HAN$ (common angle to $\triangle AMC$ and $\triangle ANH$) $\therefore \triangle AMC \parallel \triangle ANH$ (equiangular) $\therefore \angle ACB = \angle AHN$ (corresponding angles in similar triangles, $\triangle AMC \parallel \triangle ANH$, are equal) Also, $\angle AHN = \angle BHM$ (vertically opposite angles are equal) $\angle BDA = \angle ACB$ (angles at the circumference on the same arc AB are equal) Now, In $\triangle BMD$ and $\triangle BMH$ $\angle BDA = \angle BHM$ (from above, i.e. $\angle BDA = \angle ACB = \angle AHN = \angle BHM$) $\angle BMD = \angle AMC = 90^\circ$ (vertically opposite angles equal) $\angle BMH + \angle BMD = 180^\circ$ (angle sum of straight angle, $\angle BMH + 90^\circ = 180^\circ$ $\angle AMD$, is 180°) $\angle BMH = 90^\circ$ $\therefore \angle BMD = \angle BMH = 90^\circ$ MB is common $\therefore \triangle BMD \equiv \triangle BMH$ (AAS) $\therefore HM = MD$ (corresponding sides of congruent triangles, $\triangle BMD \equiv \triangle BMH$, are equal) Note: There are other solutions that were accepted as well.	3 marks for complete correct proof with correct reasoning 2 marks for substantial working with correct reasoning that could lead to a correct proof with only a minor error 1 mark for some substantial working with correct reasoning that could lead to a correct proof	

E2	(c) When $n = 1$, $LHS = F_0 = 2^{(2^0)} + 1 = 3$ $RHS = F_1 - 2 = 2^{(2^1)} + 1 - 2 = 3$ $\therefore$ Statement is true when $n = 1$. Assume the statement is true for $n = k$, some fixed positive integer. i.e. $F_0 \times F_1 \times F_2 \times \dots \times F_{k-1} = F_k - 2$ When $n = k + 1$, $LHS = F_0 \times F_1 \times F_2 \times \dots \times F_{n-1}$ $= F_0 \times F_1 \times F_2 \times \dots \times F_k$ $= F_0 \times F_1 \times F_2 \times \dots \times F_{k-1} \times F_k$ $= (F_k - 2) \times F_k \quad \text{by assumption}$ $= (F_k)^2 - 2F_k$ $= (2^{2^k} + 1)^2 - 2(2^{2^k} + 1)$ $= 2^{2 \times 2^k} + 2 \times 2^{2^k} + 1 - 2 \times 2^{2^k} - 2$ $= 2^{2^{k+1}} + 1 - 2$ $= (2^{2^n} + 1) - 2$ $= F_n - 2 \quad \text{as required}$ If statement is true for $n = k$, it has been proved true for $n = k + 1$. Since true for $n = 1$, then proved true for $n = 2, 3, 4, \dots$	4 marks for complete correct solution 3 marks for substantial working that could lead to a complete correct solution with only one error 2 marks for substantial working that could lead to a correct solution after correctly proving true for $n = 1$ with more than one error or an incomplete solution 1 mark for correctly proving true for $n = 1$ 2 marks for complete correct show
E7	(d)(i) Let r be the radius of a typical slice $\therefore r + 1 = x \rightarrow r = x - 1$ Now, $\Delta V = \pi r^{2h} = \pi (x - 1)^2 \Delta y$	1 mark for correct radius
E7	(ii) When $x = 1, y = 2 - 1 = 1$ $\therefore V = \lim_{\Delta y \rightarrow 0} \sum_{y=0}^1 \pi (x - 1)^2 \Delta y$ Now $-y = x^2 - 2x$ $\therefore 1 - y = x^2 - 2x + 1$ $\therefore 1 - y = (x - 1)^2$ hence $\therefore V = \lim_{\Delta y \rightarrow 0} \sum_{y=0}^1 \pi (1 - y) \Delta y$	2 marks for complete correct show 1 mark for substantial correct working that could lead to a correct show
E7	(iii) $V = \pi \int_0^1 1 - y \, dy$ $= \pi \left[y - \frac{y^2}{2} \right]_0^1$ $= \frac{\pi}{2} u^3$	2 marks for complete correct solution 1 mark for substantial correct working that could lead to a correct solution